

»» Chapter XI ««

COMPLEX DIFFERENTIAL COEFFICIENTS

PREFATORY REMARKS

In this chapter we shall consider the notion of the *complex differential coefficients* of a complex-valued function whose domain is a neighborhood in $\mathbf{C}$. Their use, which is universal today in many modern theories allied with complex function theory, such as the theories of quasi-conformal maps and pseudoanalytic functions, yields very elegant treatments of questions of a geometric character and of the theory of harmonic functions of two real variables.

It will be remarked that we have delayed the study of *conformality* properties of analytic functions up to this point. Our reason is based on two considerations. The first is the severity of the time limitations imposed on a beginning course in complex function theory. The second is the desire to avoid the early introduction of a subject whose further pursuit in a serious way is best continued with an adequate background in areas such as real analysis, differential geometry, and partial differential equations, which are not likely to be available to the student beginning complex function theory.

For work on quasi-conformal maps we cite [69] and [73].

1. COMPLEX DIFFERENTIAL COEFFICIENTS

Light is cast on the approach of the present section when we keep in mind Exs. 3.6 and 3.7, Ch. X, which furnish the linear background for the

infinitesimal formulation to be made. Let $f: U \rightarrow \mathbb{C}$ where $U(\subset \mathbb{C})$ is a neighborhood of α . Let $a = \operatorname{Re} \alpha$, $b = \operatorname{Im} \alpha$. We use the canonical map σ and introduce $F = f[\sigma(x, y)]$ with domain $\sigma^{-1}(U)$. We are led at once to the notion of complex differential coefficients when we *translate* the differentiability of F at (a, b) into a corresponding condition on f . Indeed, F is differentiable at (a, b) if and only if there exist $A, B \in \mathbb{C}$, $\varepsilon: U \rightarrow \mathbb{C}$, $\varepsilon(\alpha) = 0$, ε continuous at α such that

$$f(z) - f(\alpha) = A(z - \alpha) + \overline{B(z - \alpha)} + \varepsilon(z) |z - \alpha|, \quad (1.1)$$

$z \in U$. Further at most one (A, B, ε) satisfies the stated conditions. We may proceed autonomously without reference to the $\mathbb{R}^2$ theory. We say that f is *differentiable at α* provided that an ordered triple (A, B, ε) satisfying the stated conditions exists. When f is differentiable at α we term A the *complex differential coefficient of f with respect to z at α* and denote it $f_z(\alpha)$ (conceding to the archaic) and we term B the *complex differential coefficient of f with respect to $\bar{z}$ at α* and denote it $f_{\bar{z}}(\alpha)$ (again conceding). For f differentiable at α we have

$$f_z(\alpha) = \frac{1}{2}[F_x(a, b) - i F_y(a, b)]$$

and

$$f_{\bar{z}}(\alpha) = \frac{1}{2}[F_x(a, b) + i F_y(a, b)].$$

These equations are commonly used to introduce $f_z(\alpha)$ and $f_{\bar{z}}(\alpha)$. It is true that such a formal definition is available when the existence of $F_x(a, b)$ and $F_y(a, b)$ is known but not the differentiability of F at (a, b) . But the important and natural case is that where *differentiability* is present. Indeed, without such a hypothesis the symbolism so introduced would be rendered sterile. For our purposes the hypothesis of differentiability is no drawback.

EXERCISES

1.1. With U and α as above $f: U \rightarrow \mathbb{C}$ possesses a derivative at α if and only if f is differentiable at α and $f_z(\alpha) = 0$ (Cauchy-Riemann condition).

1.2. With U and α as above $f: U \rightarrow \mathbb{C}$ is differentiable at α if and only if there exist $A, B: U \rightarrow \mathbb{C}$, each continuous at α , such that

$$f(z) - f(\alpha) = A(z)(z - \alpha) + B(\bar{z})\overline{(z - \alpha)},$$

$z \in U$. Show that A and B are not uniquely determined in the situation of differentiability but that, at all events, $A(\alpha) = f_z(\alpha)$, $B(\alpha) = f_{\bar{z}}(\alpha)$.

1.3. Develop the elementary rules of computation for the complex differential coefficients of differentiable functions (sum, product, composition).

1.4. Given g analytic on a simply connected region $A \subset \mathbb{C}$, show that the solutions of $f_z = g$, f differentiable on (standard usage) A , consist of the functions of the form $G + \bar{h}$, G a primitive of g , h analytic on A . Consider the corresponding question for $f_{\bar{z}} = g$.

2. CONFORMALITY

For a linear function $az + b\bar{z} + c$ nondegeneracy occurs if and only if $|a|^2 - |b|^2 \neq 0$. By *nondegeneracy* we mean that the linear function in question is a univalent map of $\mathbb{C}$ onto itself. Nondegeneracy implies the existence of a linear inverse (cf. Ex. 3.7, Ch. X). We turn to f differentiable at α in the sense of the preceding section. In order to study the geometric questions before us without being burdened by pathological situations of great complexity we incorporate the above condition on the coefficients of the linear theory into the local study of f and assume, save explicit mention to the contrary, that

$$|f_z(\alpha)|^2 - |f_{\bar{z}}(\alpha)|^2 \neq 0. \quad (2.1)$$

The pertinence of (2.1) as a guarantor will become clear in the course of the exposition.

Suppose that $g: [0, c] \rightarrow U$, $c > 0$, is continuous at 0 and satisfies (1) $g(0) = \alpha$, (2) $g(t) \neq \alpha$, t small and positive, (3) $\lim_0 \text{sg}[g(t) - \alpha]$ exists. Here U is the domain of f . We denote the limit, which has modulus 1, by u and introduce l_g , the g -tangent ray, as the ray with initial point α and unit direction number u . What can be said about the action of f on g ? We see that

$$\begin{aligned} \lim_0 \text{sg}[f \circ g(t) - f(\alpha)] &= \text{sg}[f_z(\alpha) u + f_{\bar{z}}(\alpha) \bar{u}] \\ &= u \text{sg}[f_z(\alpha) + f_{\bar{z}}(\alpha) \bar{u}^2]. \end{aligned}$$

The role of the nondegeneracy assumption (2.1) becomes apparent now. We say that f is *conformal at α* provided that any two given admitted g , say g and h , satisfy the angle-preserving condition: there exists a rigid motion T such that $l_{f \circ g} = T \circ l_g$, $l_{f \circ h} = T \circ l_h$. From the results of §3, Ch. X, on angles we see that this is the case if and only if there exists a ray with initial point 0 containing the points $f_z(\alpha) + f_{\bar{z}}(\alpha) w$, $|w| = 1$. This condition holds if and only if $f_{\bar{z}}(\alpha) = 0$. Indeed, when $f_{\bar{z}}(\alpha) \neq 0$, the set in question is the circumference of a circle with positive radius.

Anticonformality is similarly introduced, T being replaced by an isometry of $\mathbb{C}$ onto itself not a rigid motion. It is immediate that a necessary and sufficient condition for anticonformality is that $f_z(\alpha) = 0$.

EXERCISE

2.1. When g satisfies (1) and $g'(0)$ exists and is not 0, then (2) is assured and $u = \text{sg}[g'(0)]$. In this case the tangent ray agrees with the classical notion.

Isotropic magnification (Streckentreue). We continue with another geometric question. Supposing that f is differentiable at α but does not necessarily satisfy (2.1), we ask for the behavior of the ratio

$$\left| \frac{f(z) - f(\alpha)}{z - \alpha} \right| \quad (2.2)$$

when z is near α . On considering the behavior of (2.2) composed with $\alpha + ut$, t small not zero, u being a given number of modulus 1, we see that a necessary condition for (2.2) to possess a finite limit at α is that

$$|f_z(\alpha) + f_{\bar{z}}(\alpha) \bar{u}^2| \quad (2.3)$$

be independent of u . If $f_z(\alpha) \neq 0$, the points $f_z(\alpha) + f_{\bar{z}}(\alpha) \bar{u}^2$ fill a circumference of positive radius. Since the points have constant modulus, $f_z(\alpha) = 0$. We conclude that a necessary condition for (2.2) to possess a finite limit at α is that

$$f_z(\alpha) f_{\bar{z}}(\alpha) = 0. \quad (2.4)$$

The converse is immediate. We may formulate the question somewhat differently when f satisfies (2.1). We may ask under what circumstances the function μ defined on $[\Delta(\alpha; r) - \{\alpha\}]^2$,[†] r small and positive, by

$$\mu(z, w) = \left| \frac{f(w) - f(\alpha)}{f(z) - f(\alpha)} \right| \left| \frac{w - \alpha}{z - \alpha} \right|,$$

satisfies

$$\lim_{(\alpha, \alpha)} \mu = 1. \quad (2.5)$$

We say that f fulfilling this condition is *isotropic* at α (or is “streckentreu” = length true). It is immediate that under the nondegeneracy assumption (2.1) the condition (2.4) implies (2.5). Subject to (2.1) the condition (2.5) implies that (2.3) is independent of u and hence that (2.4) holds.[‡]

EXERCISES

2.2. Given f analytic at α , $1 < n(\alpha; f) < +\infty$. Make precise the statement “ f multiplies angles at α by $n(\alpha; f)$ ” and establish the suitably formulated assertion. What can be said about lengths?

2.3. Let $f: A \rightarrow \mathbb{C}$ where A is a region in $\mathbb{C}$. It is supposed that f is differentiable on

[†] The set $A \times A$ is denoted A^2 .

[‡] Consider $w = \alpha + vt$, $z = \alpha + ut$, t small $\neq 0$, $|u| = |v| = 1$.

4. It is trivial that if f satisfies the nondegeneracy condition (2.1) at each point of A , then f is conformal at each point of A if and only if f is analytic on A . Show that if f_z and $f_{\bar{z}}$ are continuous on A and do not take the value 0 simultaneously, then the limit of (2.2) exists and is finite for each $\alpha \in A$ if and only if either f is analytic on A or $\bar{f}$ is analytic on A . When this is so, f fulfills the nondegeneracy condition at each point of A . What happens to the proof when the continuity assumption on f_z and $f_{\bar{z}}$ is omitted?

2.4. Given that $f: \Delta(a; r) \rightarrow \mathbb{C}$ is differentiable at each point of the disk $\Delta(a; r)$, that f_z and $f_{\bar{z}}$ are continuous on $\Delta(a; r)$, and that $f_{\bar{z}}(a) = 0$. Let

$$\varphi(z, w) = \frac{f(z) - f(w)}{z - w}$$

with domain

$$[\Delta(a; r)]^2 \cap \{(z, w): z \neq w\}.$$

Then $\lim_{(a, a)} \varphi = f'(a)$.

3. COMPLEX DIFFERENTIAL COEFFICIENTS AND HARMONIC FUNCTIONS

We shall see that the notion of harmonicity in the case of two real variables may be formulated very simply in terms of complex differential coefficients. It is convenient and harmless to apply the term *harmonic* to functions whose *associates* are *harmonic* in the sense of Ch. IV. Suppose that u is harmonic in the present sense on O open. We know that in some neighborhood of a point $a \in O$, $u(z) = f(z) + \overline{f(z)}$, where f is analytic at a . We conclude that u , which is differentiable on O , satisfies $u_z(w) = f'(w)$, w near a . Hence u_z is analytic on O and consequently

$$(u_z)_z = 0, \tag{3.1}$$

an equality that is just the Laplace equation rendered in complex form up to a harmless factor, $\frac{1}{4}$. Of course, $(u_z)_z = 0$ also.

Suppose now that with O as above $u: O \rightarrow \mathbb{R}$, u and u_z are differentiable on O , and (3.1) holds. Then $u_z = f$, f analytic on O . Locally f has a primitive. On $\Delta(a; r) \subset O$ we have $u(z) = F(z) + \overline{G(z)}$ where F and G are analytic on $\Delta(a; r)$. Clearly $u(z) = \operatorname{Re}[F(z) + G(z)]$. We conclude that u is harmonic in $\Delta(a; r)$ and hence on O .

Thanks to these facts we may obtain useful information concerning a function u harmonic on an annulus

$$A = \{r < |z| < s\}, \quad 0 \leq r < s \leq +\infty.$$

Let c denote the Laurent coefficient of index -1 of the analytic function u_z .

Let g be a primitive of $u_z - (c/z)$. Then

$$(u - 2c \log |z| - g)_z = 0,$$

and hence

$$u = 2c \log |z| + g + \bar{h}, \quad (3.2)$$

where h is analytic on A . Taking real parts in (3.2) we have the representation

$$u(z) = d \log |z| + \operatorname{Re} f(z), \quad (3.3)$$

$z \in A$, where $d \in \mathbf{R}$ and f is analytic on A . This result will be useful to us in Ch. XIII.

EXERCISES

3.1. Show that in (3.3) d is determined and f is determined up to a pure imaginary constant. In fact, $d = 2c$.

3.2. Given u harmonic on a simply connected region $A \subset \mathbf{C}$, show with the aid of u_z that $u = \operatorname{Re} f$ for some f analytic on A . Conclude that there exists g analytic on A such that $\log |g| = u$ and show that g is determined up to a constant factor taking a value of modulus 1.

3.3. Let u and v each be harmonic on a region $A \subset \mathbf{C}$ and suppose that $u + iv$ is analytic at some point of A . Show that $u + iv$ is analytic on A . Specialize to the case where v is the constant 0 and for some c the set $\{u(z) = c\}$ is a neighborhood of some point of A .

3.4. Let A and B be regions in $\mathbf{C}$. Suppose that $\varphi: A \rightarrow B$ and that

$$\{u \circ \varphi: u \text{ harmonic on } B\}$$

is the set of functions harmonic on A . Show that φ is a univalent map of A onto B and that either φ is analytic on A or $\bar{\varphi}$ is. Hint: Calculate $(f \circ \varphi)_{z\bar{z}}$ for f analytic on B and derive the conclusion that either φ is analytic at each point of A or $\bar{\varphi}$ is.